TASK SPECIFICATION: P5-06 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

Interactive 30-Spot Parking Grid UI

Git Branch: `feature/P5-06-map-ui` · Target Pull Request: feature/P5-06-map-ui

Core Objective & Technical Scope:
Visual parking map rendering all 30 campus spots with live status badges, zone filters, and spot detail dialogs.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-06-map-ui`
Upstream Dependencies	P3-05, P5-02		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build SpotCard and ParkingGrid components with color-coded status badges.
2. Implement zone filter tabs (All, North Zone, South Zone, Disabled).
3. Build SpotDetailModal displaying spot telemetry and reservation button.
4. Verify responsiveness and touch targets on mobile viewports.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/parking/map/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Visual parity with master design showcase in both dark and light modes.
- Grid renders smoothly with zero layout jumps upon status changes.

VERIFICATION & TEST COVERAGE STRATEGY

- ParkingGrid component tests
- SpotCard test
- Playwright map interaction